Multivariable Logistic Regression Analysis of Diabetes and Modifiable Risk Factors for Heart Disease: 2022 BRFSS Data

Claire Xu

Abstract

Heart disease remains one of the leading causes of morbidity and mortality in the United States, making the identification of major risk factors an important public health objective. This study used data from the 2022 Behavioral Risk Factor Surveillance System (BRFSS) to examine the association between diabetes and heart disease while accounting for demographic and behavioral risk factors. The full dataset included 246,022 respondents. The primary analysis used a complete-case sample of 244,039 participants after excluding 1,983 respondents with missing diabetes values created during the recoding of gestational diabetes.

Multivariable logistic regression was used to estimate crude and adjusted odds ratios (ORs), along with 95% confidence intervals (CIs). Additional analyses examined physical activity, smoking, self-reported general health, and whether the association between smoking and heart disease differed by age. Model performance was evaluated using the receiver operating characteristic (ROC) curve, with the area under the curve (AUC) used to summarize the model's ability to correctly distinguish between participants with and without heart disease. Internal validation was performed using bootstrap resampling to evaluate how well the model would generalize to similar populations. Diabetes was independently associated with more than twice the odds of heart disease (adjusted OR = 2.339, 95% CI: 2.247–2.435). A sensitivity analysis that included respondents with gestational diabetes produced nearly identical results (adjusted OR = 2.340, 95% CI: 2.248–2.436), indicating that the primary findings were robust. Older age, male sex, current smoking, physical inactivity, and poorer self-reported general health were also significantly associated with higher odds of heart disease. The primary model showed good predictive performance (AUC = 0.786), and bootstrap validation showed minimal optimism, suggesting little evidence of overfitting. Overall, these findings reinforce diabetes as a strong independent predictor of heart disease and emphasize the importance of prevention, early detection, and management of modifiable cardiovascular risk factors.

Introduction

Cardiovascular disease (CVD), especially coronary heart disease (CHD), remains the leading cause of death in the United States, representing a major public health burden due to its high prevalence, mortality, and associated healthcare costs (Shah et al. 2025). Established risk factors for heart disease include diabetes, smoking, obesity, physical inactivity, and other behavioral and demographic characteristics (Leal et al. 2024). While these risk factors are well documented, their relative contributions can vary across populations depending on clinical, lifestyle, and socioeconomic factors. Prior BRFSS-based research found that adults with diabetes alone (without co-occurring metabolic syndrome) had higher odds of heart attack compared to those with neither diabetes nor metabolic syndrome, even after adjusting for demographic and behavioral confounders (Yang, Dye, and Li 2019).

Large surveillance systems such as the Behavioral Risk Factor Surveillance System (BRFSS) are widely used to study cardiovascular risk factors in population-based samples (Yang, Dye, and Li 2019; Leal et al. 2024; Shah et al. 2025; Yathindra et al. 2025). The BRFSS dataset has a large sample size, which helps detect small effects and compare different subgroups. However, it also has limitations, including reliance on self-reported measures for both exposures and outcomes, as well as its cross-sectional design, which prevents determination of temporal ordering. Even with these limitations, large population health datasets such as BRFSS can support prevention efforts and inform public health interventions when interpreted alongside existing literature.

This study uses 2022 BRFSS data to examine whether diabetes is associated with higher odds of heart disease after adjusting for age, sex, BMI, and smoking status. It is hypothesized that adults with a diabetes diagnosis will have higher odds of heart disease than non-diabetic adults after adjustment, consistent with prior BRFSS research linking diabetes to elevated cardiovascular risk (Yang, Dye, and Li 2019).

Three secondary questions were also examined. First, whether physical inactivity independently predicts heart disease after adjusting for BMI and age, given evidence that behavioral risk factors including physical inactivity are associated with elevated cardiovascular odds (Shah et al. 2025). Second, whether the association between smoking and heart disease differs by age group, with the expectation that this association is stronger among older adults due to cumulative exposure, building on findings that smoking-related cardiovascular risk varies meaningfully across age groups (Yathindra et al. 2025). Third, whether general health status is associated with heart disease after adjusting for behavioral and clinical factors, with the expectation that odds increase progressively from excellent to poor self-reported health.

The objective of this study is to estimate these associations using multivariable logistic regression on a large population-based sample, evaluate model performance, and interpret the results in the context of existing literature on cardiovascular risk.

Data and Methods

Data Source

This study uses the *Indicators of Heart Disease* dataset (2022), derived from the Behavioral Risk Factor Surveillance System (BRFSS), administered by the Centers for Disease Control and Prevention (Centers for Disease Control and Prevention 2022). The data were accessed through Kaggle (Pytlak 2022) and contain a cleaned subset of 41 variables and 246,022 observations selected from the original BRFSS survey, which includes over 300 variables capturing demographic characteristics, health behaviors, and chronic conditions related to cardiovascular health. The dataset contains no missing values, so all 246,022 observations were initially available for analysis.

Study Population

The study population consists of U.S. adults aged 18 and older who participated in the 2022 BRFSS telephone survey. For the primary and subgroup analysis, 1,983 respondents who reported diabetes only during pregnancy were excluded, as gestational diabetes is a distinct clinical condition from diabetes diagnoses of interest in this study and was similarly excluded in prior BRFSS-based research (Yang, Dye, and Li 2019). Respondents reporting pre-diabetes or borderline diabetes were retained and classified as non-diabetic, consistent with their absence of a confirmed diabetes diagnosis. After these exclusions, the complete-case sample for the primary analysis consisted of 244,039 respondents, including only respondents with complete data for all variables used in the analysis. A sensitivity analysis was conducted using the full sample of 246,022 respondents, with gestational diabetes coded as a separate category, to assess whether this exclusion meaningfully affected the primary results.

Outcome and Exposure Variables

The outcome variable was `HeartDisease`, a binary variable indicating whether a respondent had ever been diagnosed with a heart attack. This variable was derived from the survey item `HadHeartAttack` and recoded as Yes = 1 and No = 0.

The primary exposure variable was diabetes status, derived from the original four-level `HadDiabetes` variable. Diabetes status was recoded into a binary variable (Yes = 1, No = 0) for analysis. Respondents reporting gestational diabetes were excluded because gestational diabetes occurs during pregnancy and is clinically distinct from type 1 and type 2 diabetes. Respondents reporting pre-diabetes or borderline diabetes were classified as non-diabetic because the primary analysis focused on diagnosed diabetes. This binary coding simplified interpretation of the association between diabetes and heart disease.

Covariates

Covariates were selected based on established cardiovascular risk factors identified in the literature and the secondary research questions of interest. These included age category, sex, body mass index (BMI), smoking status, self-reported general health, physical activity, and race/ethnicity.

Age category was treated as an ordered factor with 13 levels (18–24 years through 80 years and older), using the youngest age group as the reference category. Sex was coded as a binary factor, with female as the reference category. Body mass index (BMI) was analyzed as a continuous variable and was also categorized into four groups (Underweight, Normal weight, Overweight, and Obese) using standard World Health Organization cutoffs for descriptive analyses. Smoking status was coded as a four-level factor, with never smoked as the reference category, followed by former smoker, current smoker every day, and current smoker some days. Self-reported general health was treated as an ordered five-level factor, with Excellent as the reference category and Poor representing the lowest level of health. Physical activity was coded as a binary factor, with no physical activity as the reference category. Race and ethnicity were coded as a five-level factor, with White, non-Hispanic as the reference category.

Additional variables considered in supplementary analyses included sleep duration, alcohol use, history of depressive disorder, and difficulty walking.

Statistical Analysis

Descriptive statistics were used to summarize the study population overall and stratified by heart disease status. Continuous variables were described using means and standard deviations, and categorical variables were summarized using frequencies and percentages. Univariable associations between heart disease and each covariate were assessed using chi-square tests for categorical variables and t-tests for continuous variables.

Multivariable logistic regression was used to estimate adjusted odds ratios (ORs) with 95% confidence intervals (CIs) for the association between diabetes and heart disease, adjusting for age, sex, BMI, and smoking status. The primary logistic regression model was specified as:

$$\begin{aligned}\mathrm{logit}(P(\mathrm{HeartDisease}=1)) = \beta_0 &+ \beta_1(\mathrm{HadDiabetes}) + \beta_2(\mathrm{AgeCategory}) + \beta_3(\mathrm{Sex}) \\ &+ \beta_4(\mathrm{BMI}) + \beta_5(\mathrm{SmokerStatus})\end{aligned}$$

where β_1 represents the difference in log odds of having heart disease between individuals with diabetes and those without diabetes, assuming the other predictors remain constant.

Additional multivariable models were fit to address the secondary research questions, including a model examining physical activity adjusted for BMI and age, a model examining general health status adjusted for behavioral and clinical covariates, and a model including a multiplicative interaction term between smoking status and age category to formally test effect modification.

Multicollinearity was assessed informally through pairwise correlations among covariates. Model fit was evaluated using the Hosmer-Lemeshow goodness-of-fit test, deviance and Pearson residual plots, and influence diagnostics based on Cook's distance and leverage (hat values). Model discrimination was evaluated using receiver operating characteristic (ROC) curve analysis and the area under the curve (AUC). The optimal classification threshold was identified using Youden's index to account for class imbalance in the outcome variable. Internal validation was conducted using bootstrap resampling with 200 iterations to estimate optimism in model performance. Subgroup analyses stratified by sex were conducted to assess whether the association between diabetes and heart disease differed between male and female respondents. A sensitivity analysis was conducted to evaluate the impact of excluding gestational diabetes cases on the primary effect estimate.

All analyses were conducted in R (version 4.4.1), using the `tidyverse` package for data wrangling and preparation, `gtsummary` for descriptive tables, `glm()` for logistic regression, `broom` for model output extraction, `pROC` for ROC and AUC estimation, `corrplot` for correlation visualization, and `ResourceSelection` for the Hosmer-Lemeshow test. The full analysis was written in Quarto to ensure complete reproducibility, and is available in the project's GitHub repository rather than reproduced here.

Results

Descriptive Statistics

The complete-case sample for the primary analysis included 244.039 respondents after excluding 1,983 individuals who reported gestational diabetes. Among these respondents, 13,435 (5.46%) reported a history of heart attack, indicating substantial class imbalance in the outcome variable. This type of class imbalance is common in population-based cardiovascular studies and was taken into account when evaluating model performance.

Table 1 presents descriptive statistics for the study population overall and stratified by heart disease status. Respondents with heart disease were older on average and more frequently reported diabetes, current smoking, physical inactivity, and poorer self-rated general health compared to respondents without heart disease. Mean BMI was slightly higher among respondents with heart disease (29.5 vs. 28.6), while sleep duration did not differ meaningfully between groups (7.04 versus 7.02 hours). These descriptive findings indicate clear differences in cardiovascular risk profiles between respondents with and without heart disease, supporting the need for multivariable adjustment in subsequent analyses.

Table 1: Descriptive Characteristics of Study Population, Overall and Stratified by Heart Disease Status

Characteristic	**Overall** N = 244,039[1]	**No** N = 230,660[1]	**Yes** N = 13,379[1]	**p-value**[2]
Diabetes	33,813 (14%)	29,159 (13%)	4,654 (35%)	**<0.001**
Age Category				**<0.001**
Age 18 to 24	13,066 (5.4%)	13,016 (5.6%)	50 (0.4%)	

Characteristic	Overall N = 244,039[1]	No N = 230,660[1]	Yes N = 13,379[1]	p-value[2]
Age 25 to 29	10,994 (4.5%)	10,948 (4.7%)	46 (0.3%)	
Age 30 to 34	13,137 (5.4%)	13,048 (5.7%)	89 (0.7%)	
Age 35 to 39	15,331 (6.3%)	15,178 (6.6%)	153 (1.1%)	
Age 40 to 44	16,696 (6.8%)	16,473 (7.1%)	223 (1.7%)	
Age 45 to 49	16,555 (6.8%)	16,140 (7.0%)	415 (3.1%)	
Age 50 to 54	19,711 (8.1%)	19,016 (8.2%)	695 (5.2%)	
Age 55 to 59	22,050 (9.0%)	20,942 (9.1%)	1,108 (8.3%)	
Age 60 to 64	26,573 (11%)	25,003 (11%)	1,570 (12%)	
Age 65 to 69	28,443 (12%)	26,293 (11%)	2,150 (16%)	
Age 70 to 74	25,644 (11%)	23,246 (10%)	2,398 (18%)	
Age 75 to 79	18,083 (7.4%)	16,022 (6.9%)	2,061 (15%)	
Age 80 or older	17,756 (7.3%)	15,335 (6.6%)	2,421 (18%)	
Sex				**<0.001**
Female	125,837 (52%)	120,961 (52%)	4,876 (36%)	
Male	118,202 (48%)	109,699 (48%)	8,503 (64%)	
BMI	28.7 (6.5)	28.6 (6.5)	29.5 (6.6)	**<0.001**
BMI Category				**<0.001**
Normal	69,845 (29%)	66,916 (29%)	2,929 (22%)	
Underweight	3,725 (1.5%)	3,527 (1.5%)	198 (1.5%)	
Overweight	86,883 (36%)	81,954 (36%)	4,929 (37%)	
Obese	83,586 (34%)	78,263 (34%)	5,323 (40%)	
Smoking Status				**<0.001**
Never smoked	146,475 (60%)	141,032 (61%)	5,443 (41%)	
Former smoker	68,076 (28%)	62,475 (27%)	5,601 (42%)	
Current smoker - now smokes every day	21,468 (8.8%)	19,679 (8.5%)	1,789 (13%)	
Current smoker - now smokes some days	8,020 (3.3%)	7,474 (3.2%)	546 (4.1%)	
Physical Activity	189,750 (78%)	181,272 (79%)	8,478 (63%)	**<0.001**

Characteristic	Overall N = 244,039[1]	No N = 230,660[1]	Yes N = 13,379[1]	p-value[2]
General Health				**<0.001**
Excellent	41,226 (17%)	40,639 (18%)	587 (4.4%)	
Very good	86,258 (35%)	83,800 (36%)	2,458 (18%)	
Good	76,740 (31%)	72,157 (31%)	4,583 (34%)	
Fair	30,444 (12%)	26,709 (12%)	3,735 (28%)	
Poor	9,371 (3.8%)	7,355 (3.2%)	2,016 (15%)	
Race/Ethnicity				**<0.001**
White only, Non-Hispanic	184,966 (76%)	174,250 (76%)	10,716 (80%)	
Black only, Non-Hispanic	19,183 (7.9%)	18,297 (7.9%)	886 (6.6%)	
Hispanic	22,318 (9.1%)	21,464 (9.3%)	854 (6.4%)	
Other race only, Non-Hispanic	12,049 (4.9%)	11,464 (5.0%)	585 (4.4%)	
Multiracial, Non-Hispanic	5,523 (2.3%)	5,185 (2.2%)	338 (2.5%)	
Sleep Hours	7.0 (1.4)	7.0 (1.4)	7.0 (1.9)	0.4
Alcohol Use	134,253 (55%)	128,946 (56%)	5,307 (40%)	**<0.001**
Depressive Disorder	50,037 (21%)	46,758 (20%)	3,279 (25%)	**<0.001**
Difficulty Walking	35,858 (15%)	30,748 (13%)	5,110 (38%)	**<0.001**

[1]n (%); Mean (SD)

[2]Pearson's Chi-squared test; Wilcoxon rank sum test

Univariable Associations

Initial univariable analyses showed that all covariates were significantly associated with heart disease status. The largest chi-square statistics were observed for general health (χ^2 = 9,893.9), age category (χ^2 = 7,925.1), and difficulty walking (χ^2 = 6,286.6), indicating strong associations with heart disease. Diabetes was also strongly associated with heart disease (χ^2 = 5,193.4, $p < 0.001$), supporting the primary hypothesis. Smoking status (χ^2 = 2,241.3), physical activity (χ^2 = 1,701.6), and alcohol use (χ^2 = 1,353.2) were also significantly associated with heart disease. Although race/ethnicity (χ^2 = 195.92) and depressive disorder (χ^2 = 138.0) had smaller chi-square statistics, both remained statistically significant.

Univariable logistic regression produced crude odds ratios for each covariate before adjustment for potential confounders (Table 2). Diabetes was associated with higher odds of heart disease (Crude OR = 3.69, 95% CI: 3.55–3.83). A strong age-related pattern was observed, with the odds of heart disease increasing from 1.11 among respondents aged 25–29 years to

41.21 among those aged 80 years and older, compared with the 18–24-year reference group. Self-reported general health showed a similar trend, with crude odds ratios increasing from 2.01 for respondents reporting very good health to 18.83 for those reporting poor health, compared to those in excellent health. Respondents who reported engaging in physical activity had lower odds of heart disease (OR = 0.47), indicating that physically active respondents were less likely to have heart disease. The univariable results demonstrate consistent and strong associations between heart disease and key demographic, behavioral, and clinical risk factors, supporting their inclusion in the multivariable models.

Table 2: Crude Odds Ratios from Univariable Logistic Regression

Covariate	Term	Crude OR	95% CI Lower	95% CI Upper	P-value
HadDiabetes binary	Yes	3.69	3.55	3.83	<0.001
AgeCategory	Age 25 to 29	1.11	0.74	1.66	0.606
AgeCategory	Age 30 to 34	1.78	1.26	2.53	0.001
AgeCategory	Age 35 to 39	2.64	1.93	3.67	<0.001
AgeCategory	Age 40 to 44	3.56	2.64	4.89	<0.001
AgeCategory	Age 45 to 49	6.72	5.06	9.13	<0.001
AgeCategory	Age 50 to 54	9.57	7.26	12.92	<0.001
AgeCategory	Age 55 to 59	13.77	10.48	18.53	<0.001
AgeCategory	Age 60 to 64	16.38	12.49	22.00	<0.001
AgeCategory	Age 65 to 69	21.34	16.30	28.64	<0.001
AgeCategory	Age 70 to 74	26.98	20.62	36.20	<0.001
AgeCategory	Age 75 to 79	33.59	25.65	45.10	<0.001
AgeCategory	Age 80 or older	41.21	31.49	55.30	<0.001
Sex	Male	1.93	1.86	2.00	<0.001
BMI		1.02	1.02	1.02	<0.001
BMICategory	Underweight	1.30	1.12	1.50	<0.001
BMICategory	Overweight	1.37	1.31	1.44	<0.001
BMICategory	Obese	1.55	1.48	1.62	<0.001
SmokerStatus	Former smoker	2.32	2.23	2.41	<0.001
SmokerStatus	Current smoker - now smokes every day	2.36	2.23	2.49	<0.001
SmokerStatus	Current smoker - now smokes some days	1.91	1.74	2.08	<0.001
PhysicalActivities	Yes	0.47	0.45	0.49	<0.001
GeneralHealth	Very good	2.01	1.84	2.21	<0.001
GeneralHealth	Good	4.35	4.00	4.75	<0.001
GeneralHealth	Fair	9.60	8.80	10.49	<0.001
GeneralHealth	Poor	18.83	17.14	20.72	<0.001
Race/Ethnicity	Black only, Non-Hispanic	0.79	0.73	0.84	<0.001
Race/Ethnicity	Hispanic	0.65	0.60	0.69	<0.001
Race/Ethnicity	Other race only, Non-Hispanic	0.83	0.76	0.90	<0.001

Table 2: Crude Odds Ratios from Univariable Logistic Regression

Covariate	Term	Crude OR	95% CI Lower	95% CI Upper	P-value
Race/Ethnicity	Multiracial, Non-Hispanic	1.06	0.95	1.18	0.313

Primary Model

The primary multivariable logistic regression model examined whether diabetes was associated with higher odds of heart disease after adjusting for age, sex, BMI, and smoking status. Adults with diabetes had significantly higher odds of heart disease compared to non-diabetic adults (adjusted OR = 2.34, 95% CI: 2.25–2.44, $p < 0.001$),supporting the primary hypothesis. This adjusted estimate was meaningfully lower than the crude OR of 3.69 observed in the univariable analysis, indicating that age, sex, BMI, and smoking status accounted for a portion of the unadjusted association between diabetes and heart disease.

Age remained the strongest predictor of heart disease after adjustment, with odds increasing steadily across age categories and reaching the highest level among respondents aged 80 years and older compared with the 18–24 reference group (OR = 33.40). Male sex was associated with higher odds of heart disease relative to female sex (OR = 2.00). Current smoking was also strongly associated with heart disease, with higher odds observed among daily smokers (OR = 2.72) and some-days smokers (OR = 2.49), while former smokers also showed elevated odds (OR = 1.62) compared with never smokers. BMI showed a modest positive association with heart disease, with slightly increased odds per unit increase (OR = 1.02). The primary model identified age, sex, smoking status, BMI, and diabetes as independent predictors of heart disease, with a strong and consistent age gradient observed across all models.

Figure 1 displays adjusted odds ratios and 95% confidence intervals from the primary model. After adjustment, all covariates remained significantly associated with heart disease, as their confidence intervals for diabetes, age, sex, and current smoking status did not include the null value of 1.

Secondary Models

For the second research question, physical activity remained independently associated with heart disease after adjustment for age and BMI in the multivariable logistic regression model. Respondents who reported engaging in physical activity had lower odds of heart disease compared with those who did not (OR = 0.60, 95% CI: 0.58–0.62, $p < 0.001$). BMI showed a small but statistically significant positive association with heart disease (OR = 1.02 per unit increase), and the age gradient remained strong across all categories.

For the third research question, the association between smoking status and heart disease varied across age groups in the multivariable logistic regression model, providing evidence of effect modification by age. Specifically, the interaction terms for current daily smoking in older age categories were statistically significant and indicated a weaker association between

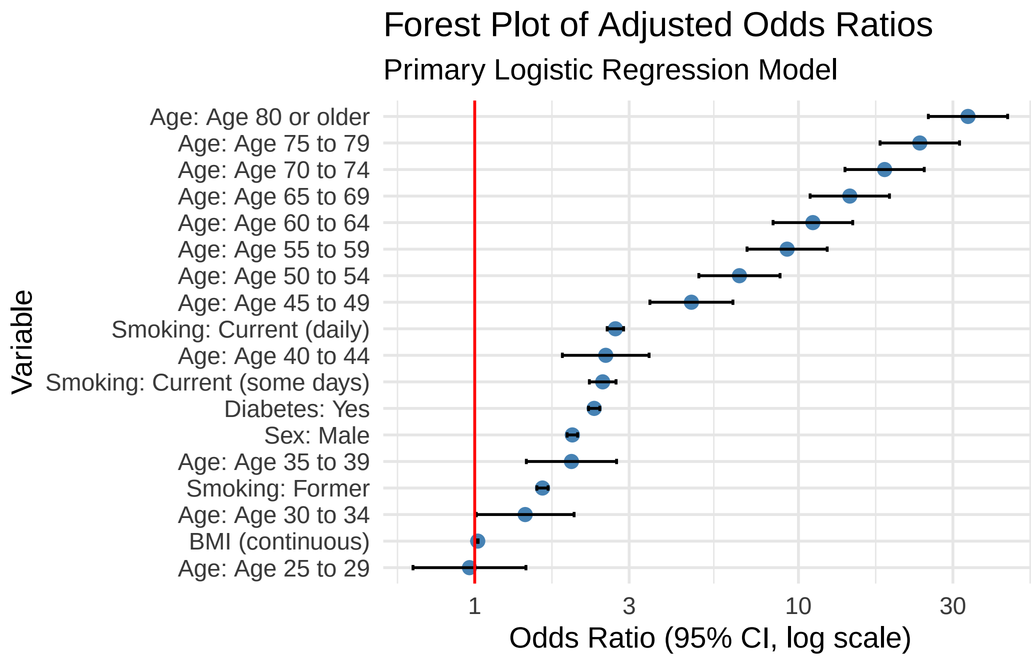


Figure 1: Adjusted Odds Ratios from Primary Multivariable Model

smoking and heart disease in older adults compared with younger adults. This was most evident in ages 70–74 (OR = 0.28, 95% CI: 0.14–0.64), 75–79 (OR = 0.21, 95% CI: 0.10–0.49), and 80 years or older (OR = 0.18, 95% CI: 0.08–0.42). These interaction effects below 1 suggest that the increase in odds of heart disease associated with current daily smoking is smaller in older age groups relative to the reference age category of 18-24.

For the fourth research question, general health status remained strongly associated with heart disease in the multivariable logistic regression model after adjustment for age, sex, BMI, smoking status, diabetes, and physical activity. Compared with excellent health, the odds of heart disease increased steadily across categories (very good: OR = 1.67; good: OR = 3.04; fair: OR = 5.66; poor: OR = 9.69; all $p < 0.001$). Although these estimates were lower than in unadjusted analyses, the association remained strong after adjustment.

Model Diagnostics

Multicollinearity among covariates was assessed using pairwise correlations. All pairwise correlations were below 0.45 (Figure 2) with the strongest observed between physical health days and difficulty walking (r = 0.43). Diabetes showed modest positive correlations with heart disease (r = 0.15) and BMI (r = 0.20), consistent with the known relationship between diabetes, obesity, and cardiovascular disease and supporting BMI's inclusion as a confounder in the primary model. No correlations were large enough to indicate concerning multicollinearity within the logistic regression framework.

The Hosmer-Lemeshow goodness-of-fit test was statistically significant ($\chi^2 = 99.41$, df = 8, $p < 0.001$), suggesting the model doesn't fit the data perfectly. This indicates some lack of fit between observed and predicted probabilities. Given the large sample size (n = 244,039),

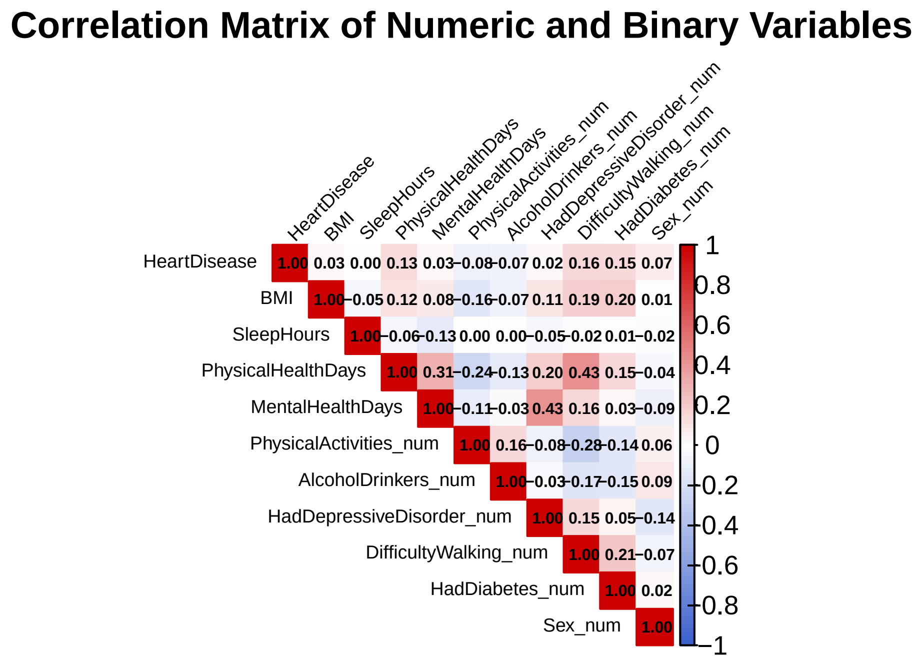
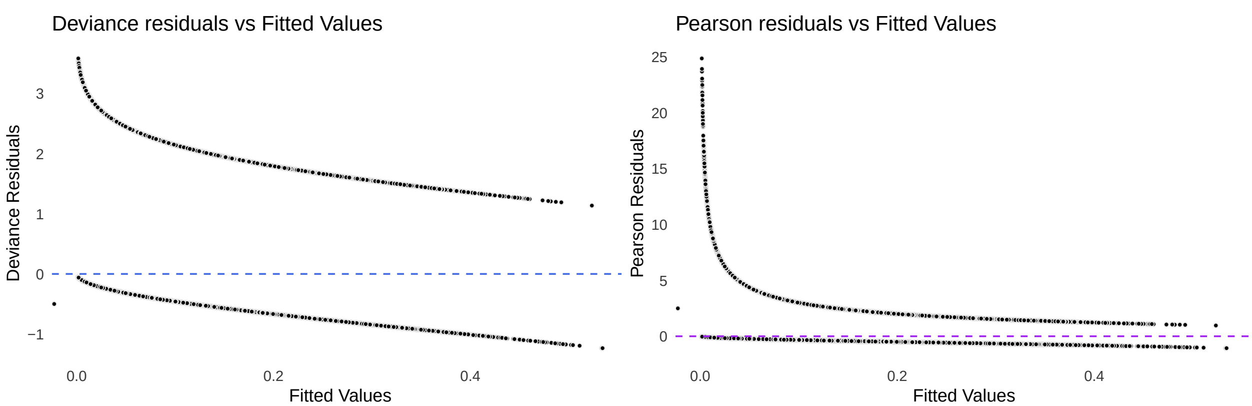


Figure 2: Correlation Matrix of Numeric and Binary Covariates

this result was expected, as even small differences between observed and predicted values can produce statistically significant results in large datasets. Deviance and Pearson residual plots (Figure 3) showed the two-band structure typical of logistic regression with a binary, imbalanced outcome, with residual magnitude decreasing as fitted probabilities increased, indicating relatively better fit among observations assigned higher predicted probabilities of heart disease.



(a) Deviance Residuals (b) Pearson Residuals

Figure 3: Deviance and Pearson Residuals vs. Fitted Values

Influence diagnostics identified 13,471 observations (5.52%) exceeding the standard Cook's distance cutoff of 4/n at around 1.64×10^{-5} (Figure 4). Refitting the model after excluding these observations resulted in coefficients shrinking toward the null (OR = 1) and odds ratios moving toward 1, indicating that these observations contributed meaningful information to the model rather than disproportionately distorting it. All observations were therefore retained in the final model. Leverage diagnostics identified 21,713 observations (8.90%) exceeding the standard hat value cutoff of 2p/n at 0.0001557128 (Figure 5). However, leverage alone does not indicate whether an observation meaningfully affects model estimates, and this role is better captured by Cook's distance.

Taken together, these diagnostic assessments provided no evidence of substantial violations that would compromise interpretation of the primary logistic regression model.

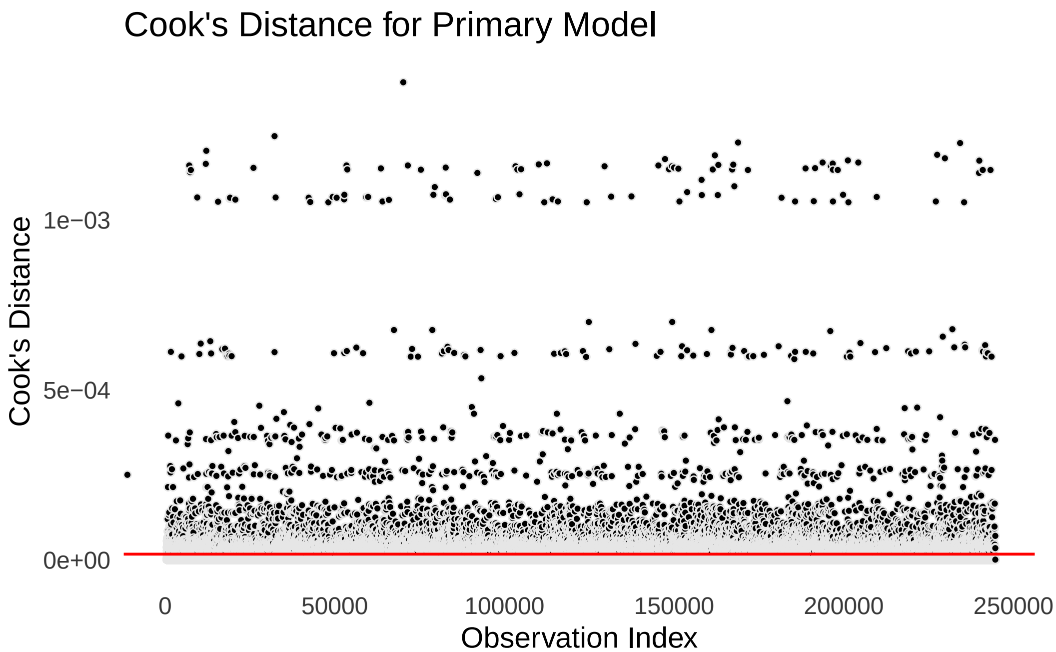


Figure 4: Cook's Distance for Primary Model

Model Performance

Model discrimination was evaluated using receiver operating characteristic (ROC) curve analysis. The primary model demonstrated good discrimination between respondents with and without heart disease (AUC = 0.786), while the full model, which included additional behavioral and clinical covariates, showed improved discrimination (AUC = 0.823) (Figure 6). Although the full model achieved higher discrimination, the primary model remained the focus of this analysis, as it was specified to directly address the primary research question regarding diabetes and heart disease after adjustment for age, sex, BMI, and smoking status.

Given the substantial class imbalance in the outcome variable, the default classification threshold (0 = No heart disease, 1 = Yes heart disease) of 0.50 was evaluated alongside an optimal threshold identified using Youden's index. At the default threshold, the model correctly identified only 1 of 13,379 heart disease cases (sensitivity = 0.0001), while correctly classifying nearly all non-cases (specificity = 0.999), confirming that the default threshold is inappropriate for this outcome prevalence (Table 3). At the Youden-optimal threshold (0.056), sensitivity improved substantially to 0.772, correctly identifying 10,334 of 13,379

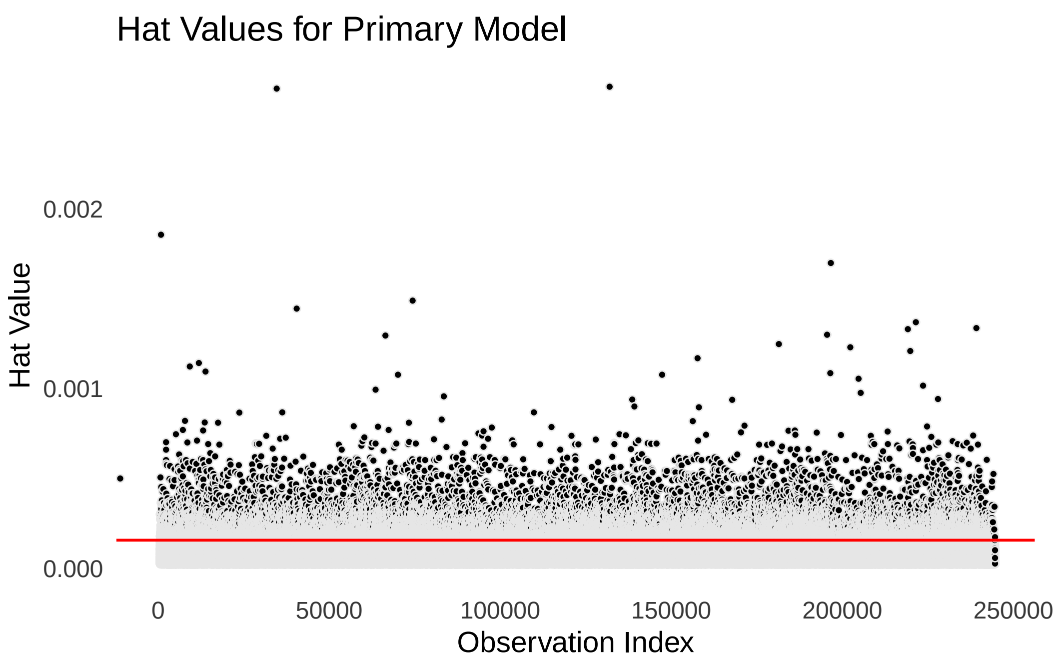


Figure 5: Hat Values for Primary Model

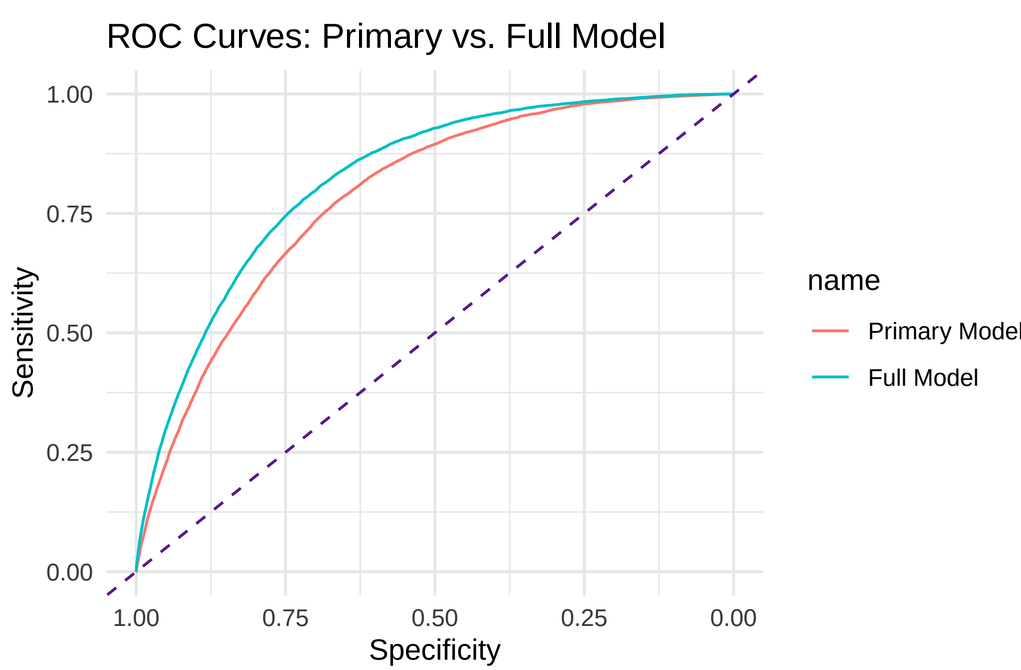


Figure 6: ROC Curves for Primary and Full Models

heart disease cases, with a corresponding decrease in specificity to 0.667 (Table 4). This threshold provides a better balance between sensitivity and specificity given the class imbalance in the data.

Table 3: Confusion Matrix Using Default Cutoff (0.50); 1 = Yes (Heart Disease), 0 = No

	0	1
0	230654	13378
1	6	1

Table 4: Confusion Matrix Using Optimal Cutoff (0.056); 1 = Yes (Heart Disease), 0 = No

	0	1
0	153822	3045
1	76838	10334

Internal validation was performed using bootstrap resampling with 200 iterations (Figure 7). The apparent AUC from the original model was around 0.786, and the bootstrap-derived mean AUC was nearly identical, resulting in an optimism estimate of 0.00016. The small optimism estimate suggests little evidence of overfitting, indicating that the model would be expected to perform similarly in comparable populations.

These findings indicate that the primary model provides reliable discrimination with little evidence of overfitting, supporting its use for evaluating the association between diabetes and heart disease.

Subgroup and Sensitivity Analyses

To assess whether the association between diabetes and heart disease differed by sex, the primary model was refit separately for female and male respondents, adjusting for age, BMI, and smoking status. The association between diabetes and heart disease was observed in both subgroups. Among female respondents, diabetes was associated with approximately 2.63 times the odds of heart disease (95% CI: 2.47–2.81) compared with non-diabetic females. Among male respondents, diabetes was associated with approximately 2.17 times the odds of heart disease (95% CI: 2.06–2.29) compared with non-diabetic males (Table 5). The point estimate was somewhat higher among females, suggesting a possibly stronger association between diabetes and heart disease in this subgroup.

Table 5: Association Between Diabetes and Heart disease Stratified by Sex

Sex	OR	CI_Lower	CI_Upper
Female	2.633	2.466	2.812
Male	2.172	2.064	2.286

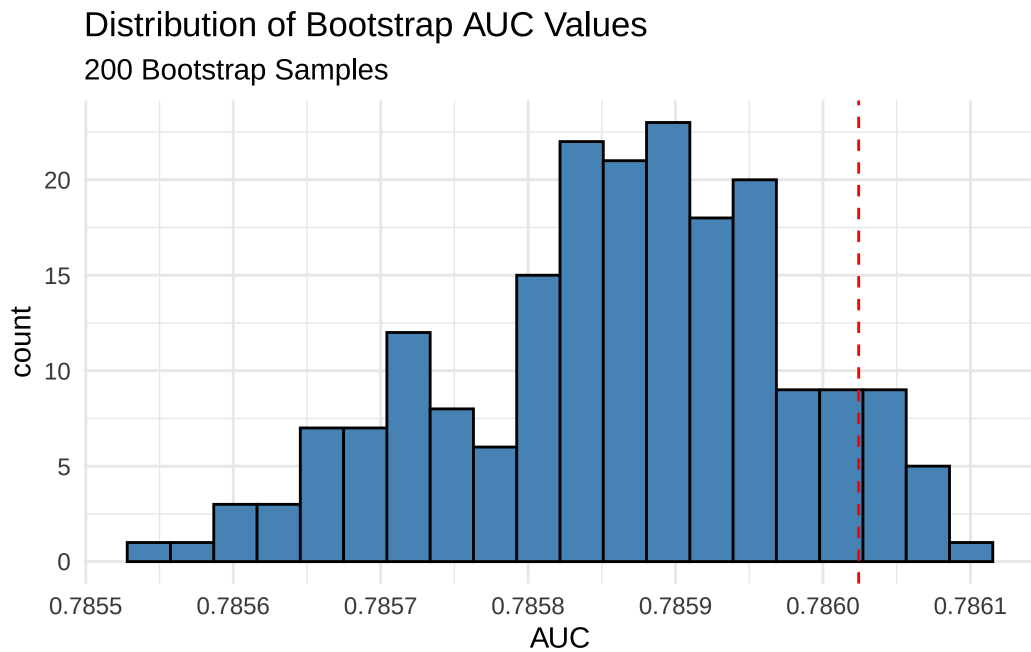


Figure 7: Bootstrap Distribution of AUC (200 Resamples)

To evaluate the impact of excluding gestational diabetes cases from the primary analysis, a sensitivity analysis was conducted using the full sample with gestational diabetes coded as a separate category. The adjusted odds ratio for diabetes was nearly identical between the primary model (OR = 2.339, 95% CI: 2.25–2.44) and the sensitivity model (OR = 2.34, 95% CI: 2.25–2.44) (Table 6), indicating that the exclusion of gestational diabetes cases did not meaningfully affect the primary findings.

Together, these subgroup and sensitivity analyses support the robustness of the primary finding that diabetes is independently associated with higher odds of heart disease, while also suggesting that this association may vary modestly by sex.

Table 6: Sensitivity Analysis: Diabetes OR with and without Gestational Diabetes

Model	OR_model	CI_Lower_model	CI_Upper_model
Primary (gestational excluded)	2.339	2.247	2.435
Sensitivity (gestational included)	2.340	2.248	2.436

Discussion

Summary of Main Findings

This study examined the association between diabetes and heart disease using 2022 BRFSS data from 244.039 U.S. adults. consistent with the primary hypothesis, adults with diabetes had significantly higher odds of heart disease compared to non-diabetic adults (adjusted OR = 2.34, 95% CI: 2.25–2.44) after adjusting for age, sex, BMI, and smoking status. This association remained stable in a sensitivity analysis including gestational diabetes cases and was observed in both female and male subgroups, though the point estimate was somewhat higher among females.

Findings from the secondary analyses were also consistent with their respective hypotheses. Physical inactivity was associated with higher odds of heart disease after adjusting for BMI and age. A significant interaction between smoking and age was observed indicating that the association between current daily smoking and heart disease varied across groups. The effect of current daily smoking on the odds of heart disease was greater in younger adults than in older adults. General health status showed a clear ordinal gradient, with progressively higher odds of heart disease as self-reported health worsened from very good to poor, relative to excellent health, even after adjustment for behavioral and clinical covariates. The primary model demonstrated good discrimination (AUC = 0.786) with minimal evidence of overfitting based on bootstrap internal validation.

Comparison to Existing Literature

The adjusted odds ratio for diabetes in the primary model was 2.34 (95% CI: 2.25–2.43), meaning that respondents with diabetes had more than twice the odds of heart disease compared with those without diabetes after adjustment for age, sex, BMI, and smoking status. This result is similar to the findings of Yang et al., who reported an adjusted odds ratio of 2.09 (95% CI: 1.72–2.54) using 2015 BRFSS data (Yang, Dye, and Li 2019). Their estimate was slightly lower, which may be due to differences in the variables included in the model. In addition to demographic factors, Yang et al. adjusted for factors such as stroke, depression, diet, and physical activity. Despite these differences, both studies found a strong positive association between diabetes and heart disease.

For physical Activity and heart disease, Shah et al. combined smoking, alcohol use, and physical inactivity into a single behavioral risk score and found that adults with two or more risk factors had 1.53 times the odds of heart attack compared with those with no risk factors (Shah et al. 2025). Although the present study examined physical activity separately, the results point in the same direction. Physically active adults had lower odds of heart disease (OR = 0.60), indicating a protective effect of physical activity. Equivalently, physical inactivity remained positively associated with heart disease risk, consistent with the findings of Shah et al.

For smoking and age, Yathindra et al. (2025) found that younger smokers aged 18 to 24 had the highest odds of angina (OR = 4.50), with the association decreasing steadily across age groups to an OR of 1.11 among adults aged 65 and older (Yathindra et al. 2025). A similar pattern was observed in this research question, where the association between smoking and heart disease became weaker in older age groups. One difference is that Yathindra et al. used angina as the outcome rather than heart disease, which may help explain their lower overall odds ratios. Despite this difference, both studies suggest that smoking has a stronger relative association with cardiovascular disease among younger adults than among older adults.

None of the studies reviewed specifically examined self-reported general health status as a predictor of heart disease using BRFSS data, which makes direct comparisons difficult.

In the present study, the odds of heart disease increased steadily as self-reported health worsened, ranging from 1.67 times higher odds among respondents reporting very good health to 9.69 times higher odds among those reporting poor health, compared with respondents reporting excellent health. This suggests that self-rated health may be a useful indicator of cardiovascular risk. Additional studies would be needed to determine whether a similar pattern is observed in other BRFSS years or national health datasets.

The age gradient observed here is consistent with published BRFSS literature. Shah et al. reported heart attack ORs increasing from 1.91 at age 25–34 to 11.22 at age 65 and older, both lower than the estimates here, which likely reflects the broader age groupings they used (Shah et al. 2025). The adjusted OR for male sex in this study (OR = 2.00 in primary model) is close to the 2.15 reported by Shah et al . and consistent with Leal et al., who found sex to be a significant predictor of CHD in 2020 BRFSS data (Shah et al. 2025; Leal et al. 2024).

Comparisons across studies should be interpreted with caution because the analyses were conducted using different BRFSS survey years (2015, 2020, 2022, 2023, and 2025), different cardiovascular outcome definitions (heart attack, angina, or coronary heart disease), and different sets of adjustment variables. Despite these differences, the overall findings were consistent, with diabetes, smoking, physical inactivity, older age, and male sex all associated with higher odds of cardiovascular disease among U.S. adults.

Limitations

Limitations should be considered when interpreting the findings of this study.

Although the multivariable model adjusted for age, sex, BMI, smoking status, and other covariates, unmeasured confounders may still influence the observed associations. Variables such as cholesterol levels, blood pressure, family history of heart disease, and medication use were not available in this dataset and could account for part of the association between diabetes and heart disease. As a result, the adjusted odds ratios reported here should not be interpreted as causal estimates, since unmeasured factors may contribute to the observed association between diabetes and heart disease.

All exposures and the outcome of heart disease were based on self-reported survey responses. Respondents were asked whether they had ever been told by a doctor that they had a heart attack, had diabetes, or engaged in physical activity, among other questions. Self-reported data are subject to recall bias and social desirability bias, and may result in misclassification of both the exposure and outcome. Individuals with undiagnosed diabetes or heart disease would be classified as not having these conditions, which could weaken the observed relationship between diabetes and heart disease and lead to an underestimate of the true association.

The BRFSS is a telephone-based survey, which may underrepresent populations without reliable phone access, including lower-income adults, those experiencing homelessness, and non-English speakers. If these groups differ systematically in their cardiovascular risk profiles, the study sample may not fully representative of the U.S. adult population, and findings may not generalize to all subgroups equally.

The BRFSS also collects information at a single point in time, so it is not possible to determine whether diabetes occurred before heart disease. Because the timing of these conditions is unknown, the observed association cannot be interpreted as a cause-and-effect relationship. For example, some individuals may have been diagnosed with diabetes only after receiving medical care for heart disease. Longitudinal studies that follow individuals over time would be needed to determine whether diabetes increases the risk of developing heart disease.

The primary exposure, diabetes status, was recoded into a binary variable by excluding 1,983 respondents who reported gestational diabetes and combining pre-diabetes with no diabetes. Although the sensitivity analysis showed that this decision did not meaningfully change the primary odds ratio (2.339 vs. 2.340), diabetes status was simplified into broad categories that may not fully reflect differences in disease severity or clinical history.

Taken together, these limitations suggest that even though the link between diabetes and heart disease is consistent across multiple sensitivity checks, it should still be interpreted as an association rather than proof of causation. It should be considered together with other cross-sectional and longitudinal studies.

Public Health Implications

Diabetes was strongly associated with heart disease in this study, with adults with diabetes having 2.34 times the odds of heart disease compared with those without diabetes after adjustment for age, sex, BMI, and smoking status. Given the high prevalence of diabetes in the United States, this finding highlights the importance of diabetes prevention, early detection, and effective disease management as part of efforts to reduce the burden of cardiovascular disease.

Several modifiable behavioral factors were also associated with heart disease. Current smoking was linked to higher odds of heart disease, and physically active adults had lower odds of heart disease than inactive adults. These findings support existing public health initiatives that encourage smoking prevention and regular physical activity as strategies for improving cardiovascular health and reducing disease risk.

The strong gradient observed between self-reported general health and heart disease suggests that a simple self-rated health measure may provide useful information about cardiovascular risk. Individuals reporting fair or poor health had substantially higher odds of heart disease compared to those reporting excellent health. Because self-rated health is easy and inexpensive to collect, it may be a useful tool for identifying populations that could benefit from additional cardiovascular risk assessment and prevention efforts.

These findings support continued efforts to improve diabetes prevention, early detection, and disease management. Public health strategies that promote healthy lifestyles and cardiovascular risk reduction among adults with diabetes may help reduce the burden of heart disease and improve population health outcomes.

Conclusion

This study examined the association between diabetes and heart disease using 2022 BRFSS data from 244,039 U.S. adults, finding that diabetes was independently associated with more than twice the odds of heart disease after adjustment for age, sex, BMI, and smoking status. This finding was robust across a sensitivity analysis including gestational diabetes cases and across sex-stratified subgroup analyses, and was consistent with prior BRFSS-based research. Secondary analyses further showed that physical inactivity, current smoking, and poorer self-reported general health were each independently associated with higher odds of heart disease, while the association between smoking and heart disease weakened with increasing age.

These findings reinforce diabetes as a key modifiable risk factor for cardiovascular disease and support continued public health emphasis on diabetes prevention, early detection, and management as strategies for reducing cardiovascular risk at the population level. While the cross-sectional design being at one point in time and reliance on self-reported data limit causal interpretation, the consistency of these findings with existing literature strengthens confidence in the observed associations. Future research using longitudinal data, which follows the same individuals over time and measures their health status at multiple time points, and more detailed clinical measures would help clarify how diabetes develops in relation to heart disease and further inform targeted prevention strategies.

References

Centers for Disease Control and Prevention. 2022. "Behavioral Risk Factor Surveillance System (BRFSS)." https://www.cdc.gov/brfss/.

Leal, Raeann, Rhonda Spencer-Hwang, W. Lawrence Beeson, Michael Paalani, and Hildemar Dos Santos. 2024. "Lifestyle Factors and Heart Health: Exploring Effect Modification Using Behavioral Risk Factor Surveillance System Survey Data." *American Journal of Lifestyle Medicine* 20 (1): 55–69. https://doi.org/10.1177/15598276241226930.

Pytlak, Kamil. 2022. "Indicators of Heart Disease Dataset." https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease.

Shah, Gulzar H., Suhail Chanar, Stuart H. Tedders, Kabita Joshi, and Kristina Harbaugh. 2025. "The Interaction of Health Behaviors and Cardiovascular Diseases: Investigating Morbidity Risks of Disparities in U.S. Adults." *Healthcare* 13 (23): 3072. https://doi.org/10.3390/healthcare13233072.

Yang, Guang-Ran, Timothy D. Dye, and Dongmei Li. 2019. "Association Between Diabetes, Metabolic Syndrome and Heart Attack in US Adults: A Cross-Sectional Analysis Using the Behavioral Risk Factor Surveillance System 2015." *BMJ Open* 9 (9): e022990. https://doi.org/10.1136/bmjopen-2018-022990.

Yathindra, Meenakshi R., Maheswari Pulluru, Amol Deokar, Faizaan Farukh Vohra, Mahima Kamble, and Bhavana Nelakuditi. 2025. "Relationship Between Smoking and Angina Across Diverse Demographic and Socioeconomic Subgroups in the United States: A Retrospective Study Using the Behavioral Risk Factor Surveillance System (BRFSS) Database." *Cureus*. https://doi.org/10.7759/cureus.39331.